VECTO 4.x Release Notes



VECTO v4.3.4-DEV (10-03-2025)

This development release addresses the issue discussed on previous workshops concerning the definition of Thermal Derating for electric motors and IEPC. The modifications implemented are the following:

1. A new definition is proposed for torque limits in derating mode and transition to derating mode (through merge request !326, which describes the new equation used).
2. A specific correction is added for IEPC recently declared overload error due to downshift with high inertia effect, allowing for those cases to run without any issue (Merge request !324).

The new derating mode definition can generate changes in VECTO results for any vehicle equipped with an electric motor or IEPC, therefore stakeholders are invited to test this development release and provide feedback before introducing the modifications in a future VECTO official release.

Issues tentatively addressed by this version (the list could change according to discussions that will follow): #761, #767, #826, #830, #850, #867, #877, #878, #886, #888, #889, #893, #939 (and clone #940, #959), #941, #942, #943, #947, #955, #956, #957, #958, #960, #962, #963, #969, #970, #971, #974 (and clone #975), #982, #983.

Bug Fixes

- EM-IEPC Thermal Derating - Tq_max and Buffer Mods (vecto/vecto!326)
- Add Driving Actions for IEPC gearshift (vecto/vecto!324)

VECTO v4.3.3 Official Release (04-03-2025)

Bug Fixes

- Check if XML element is signed (#950) (vecto/vecto!314)
- NgTankSystem optional for HEV lorries MRF XSD (vecto/vecto!315)
- fix: monitoring report for dual fuel vehicles (vecto/vecto!318)
- fix: secure XML loading against external entity injection (vecto/vecto!319)

- fix: correct interim supercap reader type (vecto/vecto!320)
- fix: parameter IDs for XSDs v2.3 and v2.6 (vecto/vecto!322)

Documentation

- Update XSD parameter IDs documentation (vecto/vecto!327)

VECTO v4.3.3 Official Release (03-03-2025)

Bug Fixes

- Check if XML element is signed (#950) (vecto/vecto!314)
- NgTankSystem optional for HEV lorries MRF XSD (vecto/vecto!315)
- fix: monitoring report for dual fuel vehicles (vecto/vecto!318)
- fix: secure XML loading against external entity injection (vecto/vecto!319)
- fix: correct interim supercap reader type (vecto/vecto!320)
- fix: parameter IDs for XSDs v2.3 and v2.6 (vecto/vecto!322)

VECTO v4.3.2-RC (06-02-2025)

Features

- New battery and supercap readers (vecto/vecto!309)

Bug Fixes

- ATShiftStrategyOptimized - No UpshiftFomL if not locked (vecto/vecto!301)
- Forbid downshift to locked gear in APT-S if it generates direct upshift condition (vecto/vecto!307)
- Add condition to write BusAuxiliaries output data in vsum (vecto/vecto!306)
- Allow old XMLs for battery and supercap in development only (vecto/vecto!311)
- Correct binding for supercap input data class; use correct xml data type in xml component reader for supercap (vecto/vecto!310)

VECTO v4.2.7 Official Release (09-01-2025)

Bug Fixes

- Track release_notes.md for release
- CodeEU #858: Convert steering pump tech (vecto/vecto!303)
- Converter Tool: ngTankSystem for dual fuel (vecto/vecto!304)
- CodeEU #836: Restrictions on IEPC gear and MaxTorqueCurve XSD attributes (vecto/vecto!302)

VECTO v4.2.6-RC (06-12-2024)

Features

- Support Gitlab issue pattern (vecto/vecto!272)
- CodeEU #854: Verify primary bus VIF hash against Job (vecto/vecto!295)
- CodeEU #833, #834, #835, #836, #837: Add 3rd amendment XSD definitions (vecto/vecto!291)
- CodeEU #838: Include Engine into v2.6 (vecto/vecto!297)

Bug Fixes

- Homogenize versions across tools (vecto/vecto!270)
- CodeEU #807: Produce same data from ADC loss map (#807) (vecto/vecto!278)
- Authors and readme metadata content (vecto/vecto!277)
- CodeEU #809: Writing engine information in MRF (#809) (vecto/vecto!275)
- CodeEU #842, #840, #841, #839, #798: Driver model: in case of an APT vehicle where the driving action is Brake… (vecto/vecto!287)
- CodeEU #812, #788: Conversion of doubles for SI (vecto/vecto!283)
- CodeEU #750, #758, #769, #816, #821, #829: Avoid wrong upshift and downshift for light SMT vehicles (vecto/vecto!290)
- CodeEU #844, #705, #530: During a coasting action (look-ahead coasting) a gear hunting occurs in the… (vecto/vecto!289)
- CodeEU #784: Override DoWriteModalResult for VTP (vecto/vecto!294)
- CodeEU #495, #642, #739: Add SMT downshift condition - DroppedSpd>DisengSpd (vecto/vecto!293)

VECTO v4.2.5 Official Release (02-10-2024)

Hot Fixes

- Missing Build.props DefineConstants (!268)
- Version 4th number read from Build.props (!269)

VECTO v4.2.3 Official Release (01-10-2024)

Enhancements

- CodeEU #799: Adapt VECTO for the new CI updates (!263)

Bug Fixes

- CodeEU #794: Added missing monitoring report file (!265)
- CodeEU #780: Update weights for bus subgroups (!264)

Vecto 4.2.2.3539 RELEASE CANDIDATE (09-09-2024)

Bug Fixes

- CodeEU-710: Hashing tool check fail with VECTO version 3330
- CodeEU-711: Hashing tool check fail with VECTO 4.1.3.3415
- CodeEU-712: VECTO VTP error
- CodeEU-754: “Failed to find operating point”; “Failed to find mechanic power for given electric power” in E2 vehicle
- CodeEU-727: Failure in simulating HEV in different VECTO versions
- CodeEU-749: Double summary for electric vehicles
- CodeEU-542: IVECO confidential : BUG REPORT : CRW LE T7D VOITH NXT 5.63
- CodeEU-663: IHPC: Failed to find operating point
- CodeEU-634: Article 10(2) issue - VIN YS2G6X20002202570
- CodeEU-671: IHPC: simulation abort due to unexpected response

Vecto 4.2.1.3469 OFFICIAL RELEASE (01-07-2024)

Features

- CodeEU-726: Build an XML converter tool for older VECTO jobs

Bug Fixes

- CodeEU-719: the six new tyre dimensions from line 126 onwards to the latest “wheels.csv” file in the VECTO repository
- CodeEU-717: VECTO-4.2.0.3448-RC - Buses AMT Gearbox Type with 1% higher C02 in primary results
- CodeEU-724: Error in Primary Bus Simulation: Object reference not set to an instance of an object
- CodeEU-716: VECTO-4.2.0.3448-RC - Buses Result Summary section missing in RLST_Customer.xml
- CodeEU-694: Primary and Completed heavybus FCV article 9 exempted hashcode mismatch.
- CodeEU-735: SMT strategy different between engineering and declaration mode
- CodeEU-736: Existing customer reports (CIF) fail validation
- CodeEU-737: Missing data from XML report

Vecto 4.2.0.3448 RELEASE CANDIDATE (10-06-2024)

Features

CodeEU-696: Double summary in CIF for vocationals and non vocational missions.

CodeEU-698: Incorporate missions RD, LH and EMS to class 16 vehicles

CodeEU-697: Re-evaluate subgroup allocations for Long Haul

CodeEU-676: Feature: Implement monitoring report

Bug fixes (1/4)

- CodeEU-471: VectoSimulationException: VF640J869RB022573
- CodeEU-462: Article10-2-issue | Order-Nr 28206354 | HEV P1 Error Mercedes-Benz and Setra Hybrid
- BusesCodeEU-326: Article10-2-issue | Order-Nr 28195581 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-348: Article10-2-issue | Order-Nr 28204519 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-457: Article10-2-issue | Order-Nr 28174000 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-458: Article10-2-issue | Order-Nr 28186528 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-459: Article10-2-issue | Order-Nr 28203057 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-545: IVECO confidential : BUG REPORT : HEV-P1 : UW18m C9 VOITH NXT CRU 48V mild hybrid
- CodeEU-346: Article10-2-issue | Order-Nr 28202338 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-352: Article10-2-issue | Order-Nr 28202130 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-394: Article10-2-issue | Order-Nr 28204065 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (2/4)

- CodeEU-433: Article10-2-issue | Order-Nr 28192321 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-451: Article10-2-issue | Order-Nr 28204280 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-452: Article10-2-issue | Order-Nr 28197394 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-453: Article10-2-issue | Order-Nr 28199435 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-454: Article10-2-issue | Order-Nr 28206982 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-655: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208126 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-660: Retarder lossmap error in completed vehicle simulation
- CodeEU-662: Generic retarder map speed range insufficient in some cases

- CodeEU-648: Error in Multistep Tool PEV/P-HEV
- CodeEU-618: PEV vehicles simulation error depending the time format
- CodeEU-700: Factor Method Generic IHPC Powermap De-normaization bug
- CodeEU-482: Article10-2-issue | Order-Nr 28203040 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-514: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208051 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-529: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28199994 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (3/4)

- CodeEU-544: IVECO confidential : BUG REPORT : HEV-P1 : CRW LE C9 VOITH NXT CRU 48V mild hybrid
- CodeEU-552: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28201759 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-556: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208176 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-557: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28209751 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-622: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28210594 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-632: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28210591 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-672: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28208841 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-673: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209179 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-674: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28211540 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (4/4)

- CodeEU-678: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28192673 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-697: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209551 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-685: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28200286 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-686: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28201178 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-687: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209679 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

- CodeEU-692: Article10-2-issue | VECTO-4.1.3 | Order-Nr 28209789 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Vecto 4.1.3.3415 OFFICIAL RELEASE (08-05-2024)

Hot fixes

- CodeEU-638: Incorrect construction payloads for group 9 non-vocational vehicle

Bug fixes

- CodeEU-615: Multistep freezes after loading VIF chassis
- CodeEU-616: Multistep tool freezes
- CodeEU-619: restore wrong exempted techs in XSD for backwards compatibility
- CodeEU-617: Results change depending the time format
- CodeEU-635: Revert multiple summary in CIF

Vecto 4.1.0.3392 RELEASE CANDIDATE (15-04-2024)

Features

- CodeEU-577: Add missing mission profiles to vocational and non-vocational

Bug fixes (1/4)

- CodeEU-331: Gear 1 DrivingActionAccelerate: Fail
- CodeEU-367: Gear 1 DrivingActionAccelerate: Fail
- CodeEU-372: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-373: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-374: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-375: ADT Error on Bus Category Primary Vehicle Simulation on VECTO
- CodeEU-393: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-439: 615 (Interurban _P32DD_ReferenceLoad) - absTime: 7129.6359 [s], distance: 53875.9765 [m], dt: 0.6388 [s], v: 0.5098 [m/s], Gear: 1 | DrivingActionAccelerate: Failed
- CodeEU-446: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-449: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-478: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 4 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 | DrivingActionAccelerate: Failed to find operating poi

Bug fixes (2/4)

- CodeEU-481: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 4 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 |
- CodeEU-488: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 4 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 | DrivingActionAccelerate: Failed to find operating poi
- CodeEU-494: Finished Run VEH-PrimaryBus_nonSmart Urban _P31SD_ReferenceLoad with ERROR: 26 (Urban _P31SD_ReferenceLoad) - absTime: 8606.6241 [s], distance: 39112.5127 [m], dt: 1.1131 [s], v: 0.1833 [m/s], Gear: 1 | DrivingActionAccelerate: Failed to fi
- CodeEU-420: Finished Run VEH-PrimaryBus_nonSmart Interurban _P32SD_ReferenceLoad with ERROR: 16
- CodeEU-573: Unhandled powertrain architecture Conventional Vehicle to calculate gradability
- CodeEU-582: unhandled powertrain architecture ConventionalVehicle to calculate gradability, 11:44:36.63,
- CodeEU-595: Inconsistency exempted vehicles “Fuel cell vehicle” vs. “FCV Article 9 exempted”
- CodeEU-501: Help regarding VTP vdri format is not up to date in v4
- CodeEU-594: inadequate validation for auxiliaries in completed vehicle XML
- CodeEU-580: Potential error in XML schema v2.4 (exempted vehicles)
- CodeEU-551: Article-10-2 XLRASF5E00G419905
- CodeEU-547: Max ICE Off timespan for buses
- CodeEU-583: Incorrect internal resistance for SuperCap used in factor method

Bug fixes (3/4)

- CodeEU-571: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28202896 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-526: VTP calculation aborted on fuel consumption map
- CodeEU-502: Error in VECTO calculation WMA10CZZ0RF022326
- CodeEU-401: Electric steering system with conventional vehicle not according to 2017/2400
- CodeEU-476: Setting Initial SoC in REESS editor leads to error
- CodeEU-533: Question about heavy lorry_IEPC_Gb×4speed: Failed to generate electric power map - at least two negative entries are required
- CodeEU-456: Clarification Documentation official Results
- CodeEU-506: Article 10(2) issue - VIN YS2P6X200R2201285
- CodeEU-507: Article 10(2) issue - VIN YS2P6X20005732399
- CodeEU-508: Article 10(2) issue - VIN S2P6X20005734199
- CodeEU-499: Object Reference not set Error
- CodeEU-546: Problem with P181 (Cooling Fan Technology) for a conventional vehicle
- CodeEU-516: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208044 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-517: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28208037 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

Bug fixes (4/4)

- CodeEU-519: Article10-2-issue | VECTO-4.0.3 | Order-Nr 10098181 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-527: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28196233 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses

- CodeEU-528: Article10-2-issue | VECTO-4.0.3 | Order-Nr 28196652 | HEV P1 Error Mercedes-Benz and Setra Hybrid Buses
- CodeEU-338: Eco-roll only, without engine stop, impact on CO2 emission
- CodeEU-475: XML files loading in VTP Job not OK; No auxiliary data and torque curve visible, Job can’t be saved in VECTO 4.0.3.3330
- CodeEU-610: VTP mode is broken due to changes in the Clutch component.

Vecto 4.0.3.3330 OFFICIAL RELEASE (13-02-2024)

Bug Fixes

- CodeEU-293: DistanceRun got an unexpected response
- CodeEU-298: Object reference not set to an instance of an object
- CodeEU-300: Full drive torque miscalculation
- CodeEU-336: Feature: Hashing tool must validate the previous step data for multi-step jobs
- CodeEU-337: Electric Citybus - ERROR with “31a-Specific” bus configuration
- CodeEU-343: Cannot simulate Primary Vehicle using vectocmd.exe
- CodeEU-387: VECTO sometimes fails to properly read Tyre data from primary vehicle xml
- CodeEU-427: Vehicle speed resulting to exceeding max gearbox speed
- CodeEU-428: Remove speed safety margin for gearbox re-engaging
- CodeEU-438: Mismatch XML schema vs Regulation (exempted vehicles)
- CodeEU-248: Bus P2 hybrid VECTO error in urban cycle
- CodeEU-278: Simulation crash when writing fuel consumption results to reports
- CodeEU-284: New error message and failed simulation obtained for P2 hybrid buses
- CodeEU-285: PEV vehicle error in routine to write the results with new vecto version 4.0.2.3275
- CodeEU-287: PEV_IEPC error message:“can only operate on SI Objects with the same unit”
- CodeEU-289: VECTO Simulation Error for bus with validated input data

Vecto 4.0.2.3275 OFFICIAL RELEASE (20-12-2023)

Hot fix

- CodeEU-273, CodeEU-274: Changes in the AMT shift strategy regarding idling speed caused simulation aborts
- CodeEU-260: regression fix handling overload buffer

Bug fixes (1/2)

- CodeEU-94: DistanceRun got an unexpected response
- CodeEU-153: DrivingActionAccelerate: Failed to find operating point after Overload
- CodeEU-158: Urban RefLoad DrivingActionAccelerate: Failed to find operating point (IEPC Wheelhub 1 measured)
- CodeEU-168: Vecto Declaration Simulation with P1-Hybrid shows multiple errors - Simulation Aborts!
- CodeEU-191: Boosting limits HEV ovc are not working
- CodeEU-202: EMS Standard Values: Continuous/Overload Torque 0 Nm

- CodeEU-203: HybridStrategy error for IHPC type 1 hybrid lorry
- CodeEU-206: VTP + PEMS test done by Renault Trucks France - VECTO tool errors most probably linked to automatic gearbox (with torque converter)
- CodeEU-211: Hybrid P1 configurations with errors
- CodeEU-215: MultiStep tool help
- CodeEU-216: NrOfGears in CIF is 1 for IEPC no matter how many gears it has
- CodeEU-220: Error manual transmission 2. gear
- CodeEU-224: Issue with PEV complete vehicle simulation (Vecto MultiStage)
- CodeEU-231: IVECO CONFIDENTIAL : hybrid buses completed simulation aborted
- CodeEU-234: Error Conventional Lorry Gear: 1C DistanceRun got an unexpected response
- CodeEU-235: Tyre error calculation buses “invalid xsi:type ‘TyreDataDeclarationType’”
- CodeEU-238: Bus VIF files not valid in Multistep VECTO
- CodeEU-243: Simulation aborted: Gear 5 Lossmap not sufficiant
- CodeEU-244: Signature validation fails for old (prior to v4) Manufacturer reports

Bug fixes (2/2)

- CodeEU-249: Fix handling gear torque limits in case of IEPC WheelHub motor and only one side is measured
- CodeEU-250: Maximum vehicle speed exceeded during pre-processing
- CodeEU-253: Replace VTP HeavyBus in Generic Vehicles with VTP Truck
- CodeEU-259: Inconsistent calculation of AverageRRC in CIF
- CodeEU-260: Handling of overload buffer in case Continuous Torque is 0Nm
- CodeEU-261: SuperCap internal resistance standard values correction
- CodeEU-263: Determination of rated power for IEPCs for MRF and CIF
- CodeEU-266: Fix Measured Speed Testcases

Vecto 4.0.1.3217 OFFICIAL RELEASE (23-10-2023)

Improvements

- VTP mode: create zip archive for input and output files

Bug Fixes

- bugfix in CIF for complete(d) buses - do not write PrimaryVehicleSubgroup element
- fix hybrid strategy: power comparison
- fix exempted vehicles do not work (error message that XML version is not supported)
- bugfix in XML schema: remove wrong PS technology entry for lorries
- bugfix for conditioning power demand for IHPC vehicles

VECTO 4.0.0.3210 Release Notes (16-10-2023)



Release Notes

First official VECTO release for the 2nd Amendment of Regulation (EU) 2017/2400

Features

- Declaration Mode
  - xEV-Heavy Lorries
  - Conventional Medium Lorries
  - xEV Medium Lorries
  - Conventional Buses
  - xEV Buses
- Dedicated user interface for simulating buses in Declaration Mode using the multistep approach: VECTOMultistep.exe
- Engineering Mode
  - xEV-Vehicles
- Updated XML-Reports and Vehicle Information Files to support new Vehicle Architectures
  - New Manufacturer Record File (MRF)
  - New Customer Information File (CIF)
  - Multistep Output (VIF):
- Updated simulation output (.vsum, .vmod)
- Updated Hashing Tool to handle new powertrain components (electric motor, battery, supercap ...)
- Supports .NET Framework 4.8 and .NET 6.0

Features - Tool modes

	Medium Lorries (XML)	Heavy Lorries (XML)	Primary buses (XML)	Interim buses (GUI)	Interim buses (XML)	Completed buses (JSON v7)	Completed buses (GUI)	Com bu (X
VECTO.exe	S	S	S			S		
VECTOcmd.exe	S	S	S			S		
VECTOMultistep.exe	S	S	S	S+E	S+E	S+E	S+E	S
Comments				function: generates updated VIF				

S: *Simulation only.*

S+E: *Simulation and editing.*

Changes

- **XML Job Files < v2.4 are no longer supported**
- **Dropped support for .NET Framework 4.5 (EOL 04/2022)**

Implementation of Declaration Mode for xEV-Vehicles

- **Generic auxiliary parametrisation**
- **Generic parametrisation of usable SOC range**
- **Generic parametrisation of Serial/Parallel Hybrid Strategy parameters**
- **Automated simulation of OVC-HEV**
  - **Charge depleting mode (CD): The propulsion energy is provided by the electric storage only**
  - **Charge sustaining mode (CS): The propulsion energy is provided by the fuel storage**

Implementation of Declaration Mode for xEV-Lorries

Generic E-PTO for PEV and S-HEV in Declaration Mode

E-PTO for PEV and S-HEV in Engineering Mode

Implementation of Declaration Mode for Buses

Factor Method for conventional and xEV-Buses

Updated bus auxiliary model